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Brussow et al.

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(54) **COOLING WATER MANAGEMENT SYSTEMS HAVING CONTINUOUS BIOFILM MONITORING AND ASSOCIATED METHODS FOR USING THE SAME**

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CPC C02F 1/38; B04C 3/06; B04C 9/00
See application file for complete search history.

(71) Applicants: **Bradley J. Brussow**, Belmont, MI (US); **Ed Kroll**, Whitehall, MI (US)

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(72) Inventors: **Bradley J. Brussow**, Belmont, MI (US); **Ed Kroll**, Whitehall, MI (US)

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(73) Assignee: **TOWER GUARD, LLC**, Rockford, MI (US)

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(*) **Notice:** Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

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(21) Appl. No.: **18/926,509**

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Primary Examiner — Bobby Ramdhanie

(74) *Attorney, Agent, or Firm* — King & Partners, PLC

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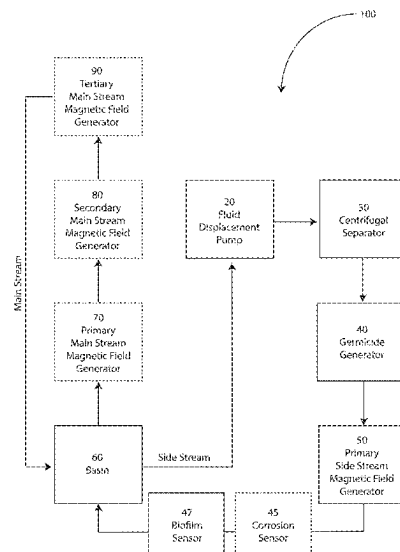
(57) **ABSTRACT**

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C02F 1/461 (2023.01)

A water cooling/recirculating management system, including: a controller; a fluid displacement pump; a centrifugal separator; a germicide generator; a primary side stream magnetic field generator; an optional continuous, real time corrosion monitoring and reporting system; and a continuous, real time biofilm monitoring and reporting system.

(52) **U.S. Cl.**
CPC **C02F 1/32** (2013.01); **C02F 1/385** (2013.01); **C02F 1/46114** (2013.01); **C02F**

3 Claims, 4 Drawing Sheets



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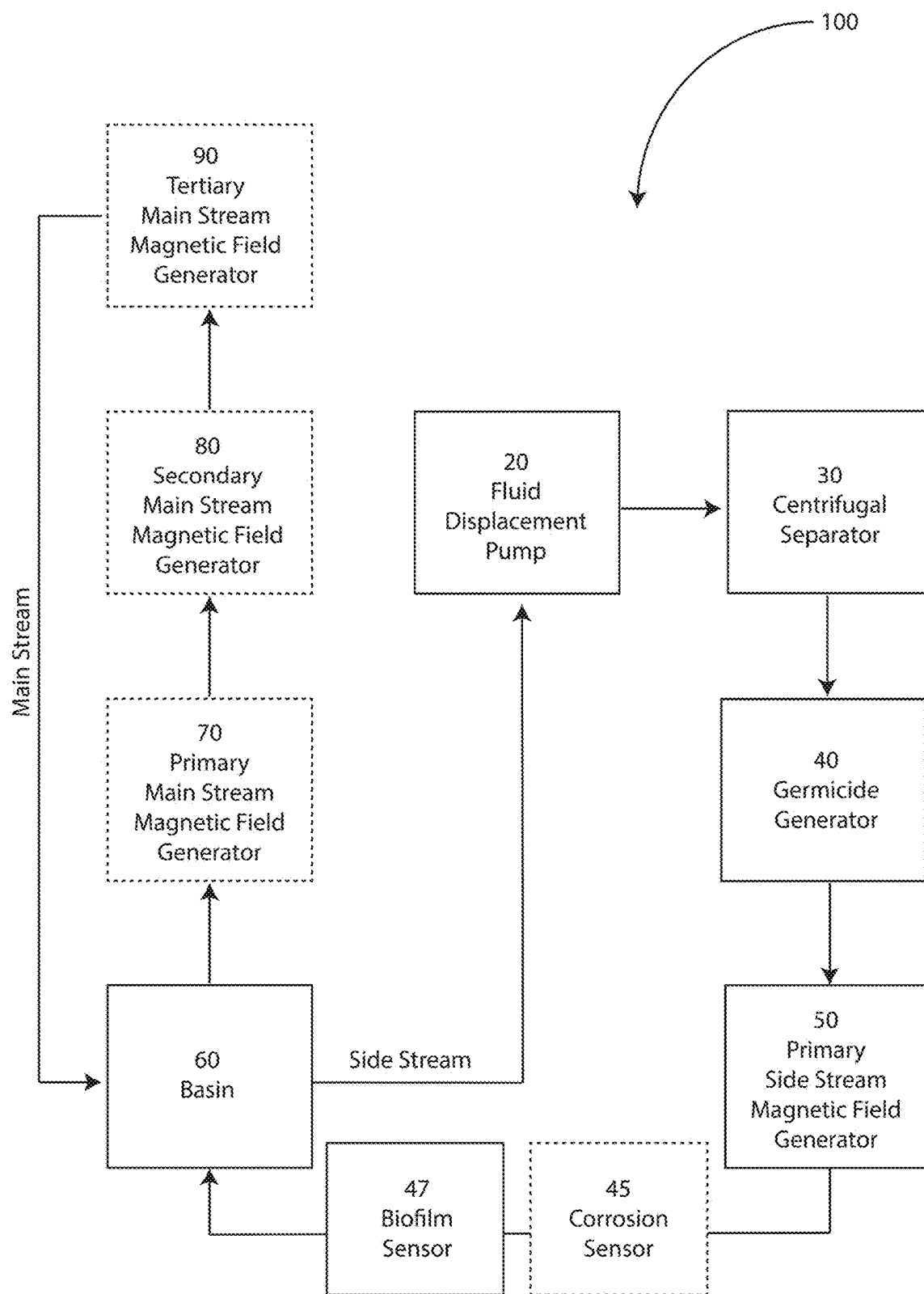


Figure 1

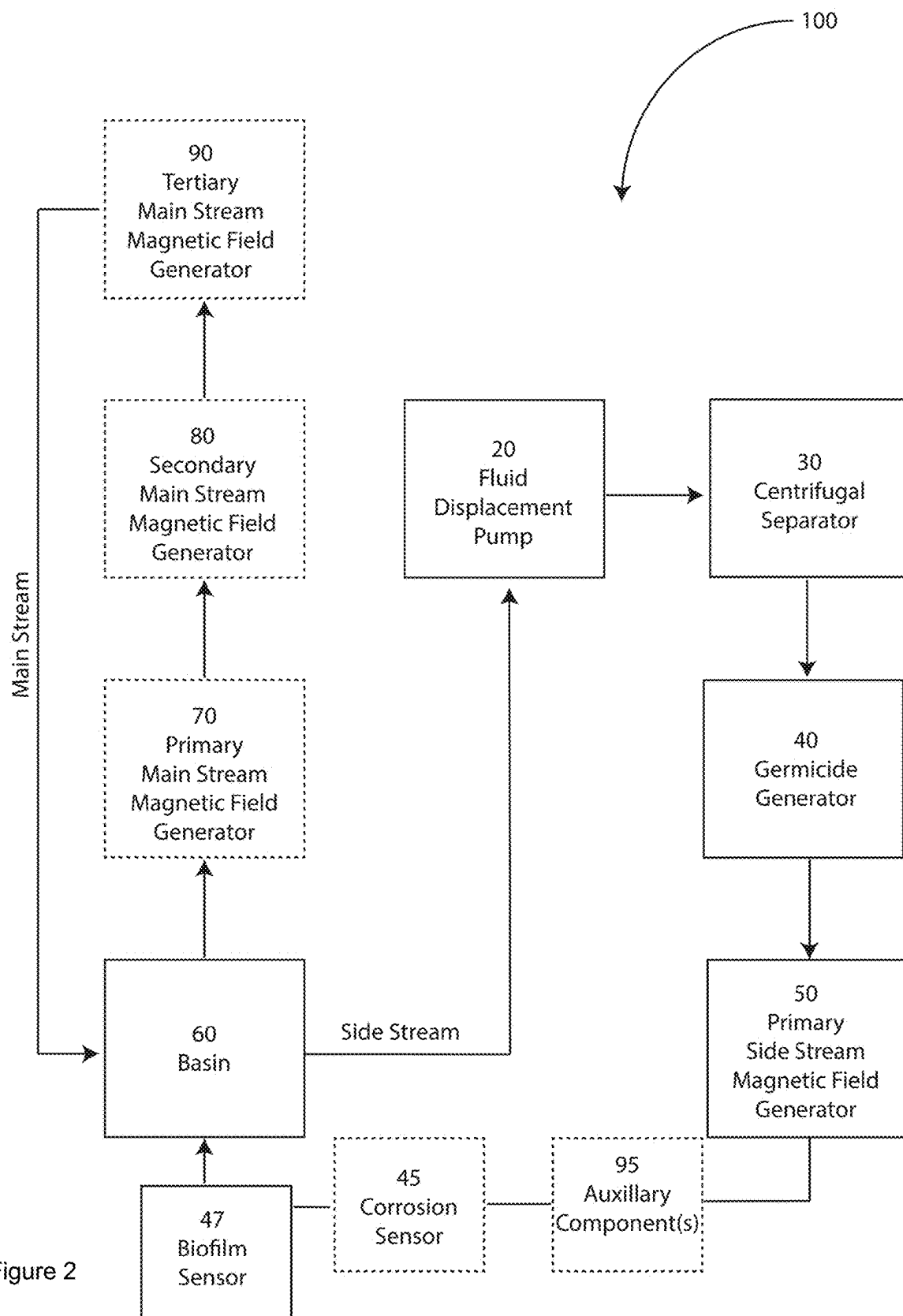


Figure 2

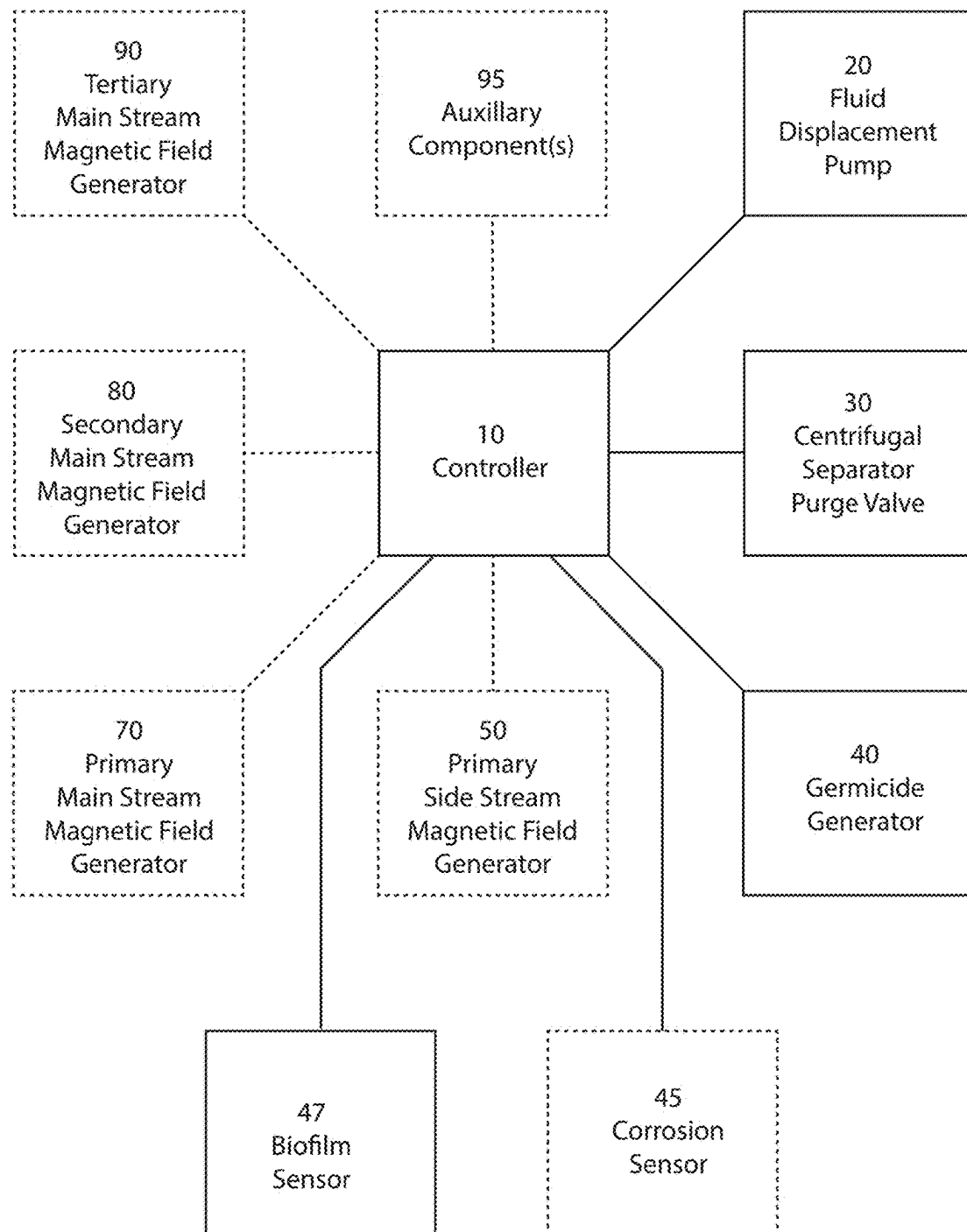
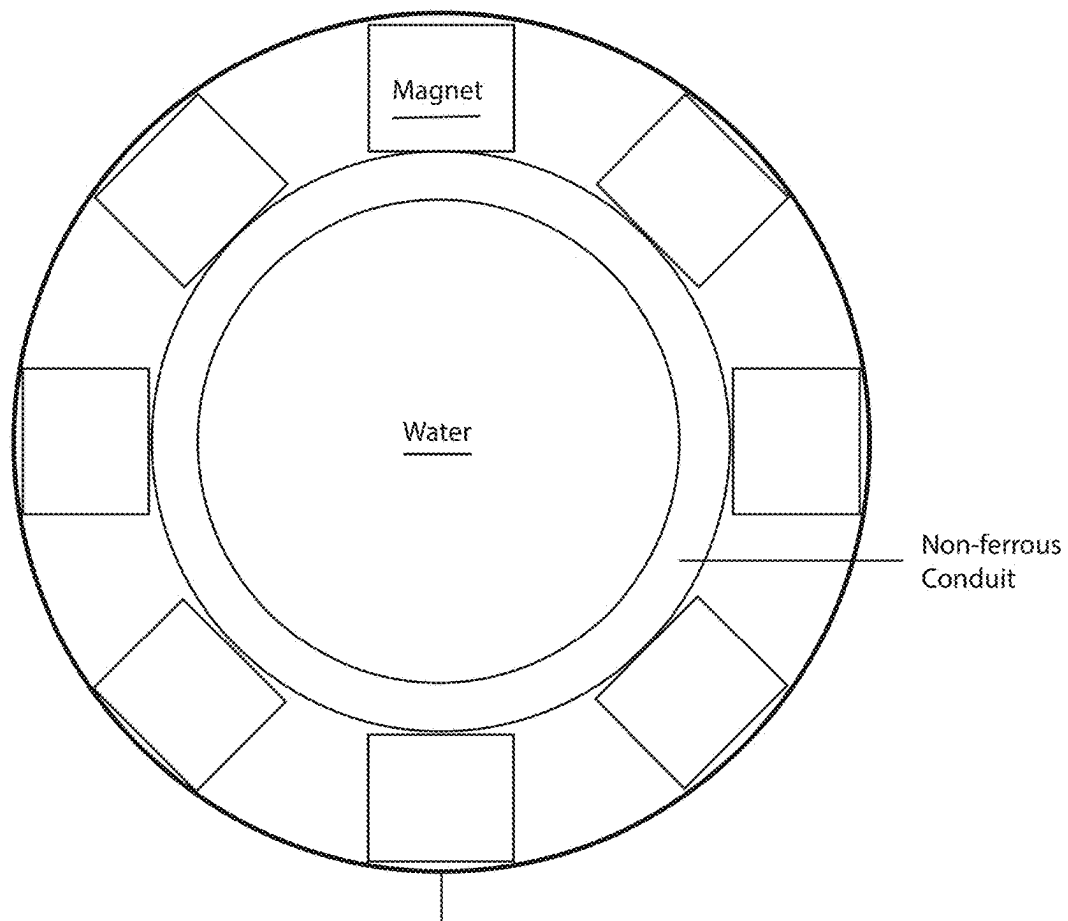


Figure 3



Steel band wrapped around all the magnets, and in direct contact with them, that focuses the otherwise-wasted (extending radially outward), magnetic flux lines into the pipe.

Figure 4

1

**COOLING WATER MANAGEMENT
SYSTEMS HAVING CONTINUOUS BIOFILM
MONITORING AND ASSOCIATED
METHODS FOR USING THE SAME**

COPYRIGHT NOTICE

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CROSS-REFERENCE TO RELATED
APPLICATIONS

This application is a continuation-in-part of U.S. application Ser. No. 17/898,741, entitled "COOLING WATER MANAGEMENT SYSTEMS AND ASSOCIATED METHODS FOR USING THE SAME," filed Aug. 30, 2022, which claims the benefit of U.S. Provisional Application Ser. No. 63/242,527, entitled "COOLING WATER MANAGEMENT SYSTEMS AND ASSOCIATED METHODS FOR USING THE SAME," filed Sep. 10, 2021-which are hereby incorporated herein by reference in their entirety, including all references cited therein.

STATEMENT REGARDING FEDERALLY
SPONSORED RESEARCH OR DEVELOPMENT

Not applicable.

REFERENCE TO A SEQUENCE LISTING

Not applicable.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates in general to water management systems having continuous biofilm monitoring and reporting, and, more particularly, to cooling and/or recirculating water management systems that eliminate the need for harsh chemicals, bleed water, and labor to maintain recirculating water systems at peak efficiency. This results in: (1) up to a 40% savings in energy consumption; (2) the elimination of chemical treatment costs; (3) the elimination of blowdown water and sewer costs; (4) the elimination of labor to clean/restore heat exchange equipment; and (5) greatly extended equipment life. The water management systems of the present invention also result in the elimination of environmental pollution from chemical bleed, and personnel protection against waterborne diseases, such as *Legionella*.

While evaporative water cooling for heat exchange systems (e.g., cooling towers, chillers, compressors, heat exchangers, HVAC systems, etcetera) and heat rejection processes is the most efficient cooling method, it is also the most aggravating to maintain because of scale, corrosion, algae, bacteria, biofilm, and sludge. The present invention eliminates the above-identified problems using physics instead of conventional, environmentally problematic chemistry by synergistically utilizing a centrifugal separator, a germicide generator, and a magnetic field generator.

2

The water management systems of the present invention also provide for continuous, real time biofilm monitoring and reporting which is a significant improvement over traditional systems that only monitor biofilms periodically (e.g., weekly, monthly, etcetera) with costly offsite water testing.

2. Background Art

Water management systems and associated components have been known in the art for years, and are the subject of a plurality of patents including, for example: U.S. Pat. No. 8,223,495 entitled "Electronic Device Cooling System," U.S. Pat. No. 5,145,585 entitled "Method and Apparatus for Treating Water in a Cooling System," U.S. Pat. No. 4,764,283 entitled "Method and Apparatus for Treating Cooling Tower Water," U.S. Pat. No. 4,519,919 entitled "Method and Apparatus for Magnetically Treating Fluids," U.S. Pat. No. 4,202,847 entitled "Apparatus and Method for Cooling Water Especially in Cooling Towers," U.S. Pat. No. 4,172,786 entitled "Ozonation of Cooling Tower Waters," U.S. Pat. No. 4,153,559 entitled "Water Treatment Device and Method for Manufacturing Same," U.S. Pat. No. 3,997,635 entitled "Method and Device for Evaporative Cooling," U.S. Pat. No. 3,669,425 entitled "Water Cooling Tower," Chinese Patent Number 201392109 entitled "Closed Type Cooling Tower," Chinese Patent Number 2906512 entitled "Energy-Saving Environmental-Protection Cooling Tower," and Chinese Patent Number 1844824 entitled "Environment-Friendly Water-Saving Cooling Tower"-all of which are hereby incorporated herein by reference in their entirety including all references cited therein.

U.S. Pat. No. 8,223,495 appears to disclose cooling systems for providing cooled air to electronic devices. The systems can include large storage tanks or waste treatment systems to improve the efficiency of the plant and reduce impact on the environment.

U.S. Pat. No. 5,145,585 appears to disclose a method for treating water in a cooling system that includes a cooling tower and heat exchanger. The method includes the steps of removing water from the main cooling water circuit, serially subjecting the water to magnetic treatment, treatment with ozone, filtering with a centrifugal separator, and then returning to the main cooling water circuit. Underflow from the centrifugal separator is filtered with a bag filter, and the filtered water returned to the magnetic treatment step. The pH of the water may be controlled by regulating the production of dry air supplied to the ozone generator.

U.S. Pat. No. 4,764,283 appears to disclose an apparatus and method of reducing agglomeration of fluids consisting of the vigorous agitation of the liquid by a double opposed vortex nozzle and the passing of the liquid through a magnetic field. One of the opposed nozzles may have ozone gas forced through it for treating sewage. The apparatus and method may also be used to treat cooling tower water.

U.S. Pat. No. 4,519,919 appears to disclose an apparatus for and method of magnetically treating flowing fluids that includes a housing into which a fluid stream is directed and in which the stream is divided into a plurality of separate flow paths. Each of the flow paths within the housing is subjected to a monopolar magnetic influence and a baffle configured to induce turbulence in the flow path to insure exposure of all of the fluid in the path to the monopolar magnetic influence. The separate flow paths are at least partially recombined to a single flow path and discharged from the housing.

U.S. Pat. No. 4,202,847 appears to disclose a water cooling tower and method which provides for both wet and dry cooling within the same vertical space and significantly reduces the discharge of water particles out of the tower. Water is deposited on one side of each of a plurality of trickler plates and flows downwardly under the influence of gravity substantially undisturbed as a low speed film. Air flows at low speed by natural draught in a substantially undisturbed path on the other side of each of the plates.

U.S. Pat. No. 4,172,786 appears to disclose a continuous ozone injection into water that circulates between a cooling tower and heat exchanger with heavy scale deposits inhibits formation of further deposits, promotes flaking of existing deposits, inhibits chemical corrosion and controls algae and bacteria.

U.S. Pat. No. 4,153,559 appears to disclose a device for the magnetic treatment of water and other liquids, having a pair of concentric tubular casings which are spaced from each other so as to form an annular treatment chamber. The inner casing contains an elongated magnet having two or more longitudinally spaced poles, and the outer casing is made of a magnetic material which serves to concentrate the magnetic lines of force within the annular chamber. The inner casing is supported within the outer casing by means of elastic, non-magnetic sleeves which are positioned over opposite ends of the inner casing and compressed between it and the inner surface of the outer casing so that the treatment chamber is rendered fluid-tight and the inner casing and magnet are prevented from shifting longitudinally with respect to the outer casing. The device is manufactured by pressing resilient end caps on opposite ends of the magnet and pressing this assembly into the inner casing, inserting the inner casing in the outer casing and then wedging the sleeves between them.

U.S. Pat. No. 3,997,635 appears to disclose a method and a device for preventing formation of mist by the discharge of moistened air from an evaporative cooler, primarily a cooling tower of the multi-layer type, comprising a contact body composed of a plurality of layers forming between themselves gaps passed by water to be cooled and atmospheric air which is moistened and heated by contact with the water in counter-current or cross-current flow. Under certain conditions as when the temperature of the atmospheric air is so low, when the moistened and heated air is discharged to the atmosphere and cooled to the low temperature of the latter, the moisture contained in the discharged air will be condensed and precipitated in the form of mist to the extent it exceeds the saturation point of the air. Such precipitation can be avoided by mixing the moistened air prior to its escape into the atmosphere with preheated dry air. The invention provides a novel method and device for producing such preheated dry air by causing part of the introduced atmospheric air to pass through gaps of the contact body shut off from the flow of water passing through the body, whereby this air is heated without change of its moisture content by heat exchange with the warm water flowing through adjacent gaps. The moist air current from the gaps wetted by the water flowing through them and the dry air current from the gaps not accessible to the water provided an air mixture which, even when discharged into the relatively cold atmospheric air and thereby cooled to the low temperature thereof, will not contain so much moisture that the saturation limit is surpassed and condensed vapor precipitated. Means are also disclosed to adjust the ratio of the gaps open to the water flow to the gaps shut off from said

flow so as to reduce the number of dry gaps in which no cooling effect is produced to the minimum necessary for avoiding mist formation.

U.S. Pat. No. 3,669,425 appears to disclose a water cooling tower which comprises a blower opening into the bottom of the tower for introducing or blowing air into the tower. A continuously walled wet decking arranged in a spiral mounted above the air inlet port and having a large number of regularly spaced projections arranged in an inclined checkerboard or step like pattern to control the counterflowing air and water in the tower. A spray tree for introducing water into the vessel is mounted above the wet decking and a mist eliminator is mounted above the spray tree. The mist eliminator is also constructed in the form of a continuously spirally wound vertical wall with adjacent surfaces of the eliminator wall being separated by a multiplicity of step like baffles.

Chinese Patent Number 201392109 appears to disclose a closed type cooling tower, which includes a fan arranged at the top of the tower, a water-catching tank arranged at the bottom of the tower, a medium inlet and a medium outlet, a heat exchanger coil arranged in the tower and a spraying water-distributing system, wherein two ends of the heat exchanger coil are respectively communicated with the medium inlet and the medium outlet; the spraying water-distributing system is connected to the water-catching tank through a cycling water pump; moreover, the spraying water-distributing system is also provided with a plurality of nozzles, wherein the heat exchanger coil is formed by a capillary; as the capillary is characterized by large heat exchange area, thin wall, good thermal conductivity, uniform heat exchange, and the like, by adopting the capillary as the heat exchanger coil, on one hand, the heat exchange effect of the cooling tower can be improved, on the other hand, under the condition of equal heat exchange requirement, the heat exchanger coil which is formed by the capillary can be designed to be smaller, thereby leading the volume of the cooling tower to be smaller.

Chinese Patent Number 2906512 appears to disclose a water-saving and environment-friendly cooling tower, which relates to a cooling device for cooling the circulating water. An air intake shutter of air cooler is symmetrically arranged on the sidewall of the cooling tower body, an air cooler is arranged close to the air intake shutter, a distributing pipe in the air cooler inside is connected with a circulating water intake pipe, an outlet pipe of the air cooler is connected with a water distributor, a lower catchment tank of the air cooler is connected with a circulating water outlet pipe by a bypass pipe. The air cooler is arranged above a water collector, the water distributor is arranged between the two air coolers. The utility model is water saving, the air cooler is used to precool the circulating water, the cooling load of the filler segment and the quantity of the evaporated water are decreased. The air becomes hot and dry when going through the air cooler and the humidity of the hot wet air, which can reduce the humidity ratio of the hot and wet air after being exchanged in the filler layer, and the water fog generated at the top of the tower is avoided or reduced, which is favorable to environmental protection.

Chinese Patent Number 1844824 appears to disclose an environment-friendly water-saving cooling tower, wherein the side wall of the tower is symmetrically distributed with wind inlet shutters of air cooler; the inner side that is near to the shutters is distributed with an air cooler; the water distributing tube at the inner side of air cooler is through to the water inlet tube of cycle water, while the water outlet tube of air cooler is through to the water distributor; the

5

lower water tank of air cooler via the branch tube is through to the reflux tube of recycle water; the air cooler is above the water collector; the water distributor is between two air coolers. The invention uses an air cooler to pre-cool the cycle water, to reduce the load, the steam amount, and the water consumption.

While the above-identified patents do appear to disclose various water management systems and associated components, their configurations remain non-desirous and/or problematic inasmuch as, among other things, none of the above-identified systems appear to provide continuous, real time biofilm monitoring and reporting.

These and other objects of the present invention will become apparent in light of the present specification, claims, and drawings.

SUMMARY OF THE INVENTION

The following presents a simplified summary in order to provide a basic understanding of some aspects of the claimed subject matter. This summary is not an extensive overview, and is not intended to identify key/critical elements or to delineate the scope of the claimed subject matter. Its purpose is to present some concepts in a simplified form as a prelude to the more detailed description that is presented later.

The present invention is directed to a water cooling/recirculating management system, comprising, consisting essentially of, and/or consisting of: (a) a controller; (b) a fluid displacement pump; (c) a centrifugal separator; (d) a germicide generator; (e) a primary side stream magnetic field generator; (f) an optional continuous, real time corrosion monitoring and reporting system; and (g) a continuous, real time biofilm monitoring and reporting system.

In a preferred embodiment of the present invention, the controller is in communication with the fluid displacement pump, the centrifugal separator purge valve, and the germicide generator.

In another preferred embodiment of the present invention, the fluid displacement pump extracts water from a main stream basin, prefilters the extracted water prior to association with the centrifugal separator, and displaces the extracted water through a closed-circuit side stream.

In yet another preferred embodiment of the present invention, the centrifugal separator includes an inlet, an outlet, and a solids collection chamber.

In one aspect of the present invention, the centrifugal separator separates solids, prevents sludge accumulation, replaces blow-down and/or bleed, removes particles down to 40 microns, contains no filter medium or moving parts, provides for the continuous removal of solids, and/or provides for a purging mechanism that releases solids collected in the collection chamber.

In accordance with a preferred embodiment of the present invention, the germicide generator comprises one or more electrodes fabricated from approximately (i.e., $\pm 3\%$) 93 percent copper and approximately (i.e., $\pm 3\%$) 7 percent silver.

In accordance with another preferred embodiment of the present invention, the germicide generator comprises at least one electrode fabricated from approximately (i.e., $\pm 3\%$) 90 percent copper and approximately (i.e., $\pm 3\%$) 10 percent silver.

In a preferred embodiment of the present invention, the germicide generator eliminates the need for chemicals, utilizes a copper/silver ionization system, and/or destroys algae, bacteria, and/or viruses.

6

In another preferred embodiment of the present invention, the germicide generator operates in an off mode, a continuous on mode, and/or a non-continuous pulse mode.

In yet another preferred embodiment of the present invention, the primary side stream magnetic field generator comprises one or more permanent ceramic magnets. In this embodiment, the permanent ceramic magnet preferably includes side-to-side polarity that is mounted to a non-ferrous pipe having the south pole facing inward toward the water.

In a preferred embodiment of the present invention, the primary side stream magnetic field generator prevents scale formation, de-scales existing surfaces and piping, inhibits both general and galvanic corrosion, and, if present, increases heat exchanger efficiency by approximately (i.e., $\pm 3\%$) 5-55 percent.

In another preferred embodiment of the present invention, the water cooling/recirculating management further comprises an ultraviolet light source, an ozone generator, and/or a cooling fan.

In yet another preferred embodiment of the present invention, the water cooling/recirculating management system further comprises a primary, secondary and/or tertiary main stream magnetic field generator.

In one embodiment of the present invention, the water cooling/recirculating management system further comprises a pressure switch to indicate the presence, flowrate, and/or velocity of the side stream water flow.

BRIEF DESCRIPTION OF THE DRAWINGS

Certain embodiments of the present invention are illustrated by the accompanying figures. It will be understood that the figures are not necessarily to scale and that details not necessary for an understanding of the invention or that render other details difficult to perceive may be omitted.

It will be further understood that the invention is not necessarily limited to the particular embodiments illustrated herein.

The invention will now be described with reference to the drawings wherein:

FIG. 1 of the drawings is schematic representation of a first embodiment of a water cooling system for use in accordance with the present invention;

FIG. 2 of the drawings is schematic representation of a second embodiment of a water cooling system for use in accordance with the present invention;

FIG. 3 of the drawings is schematic representation of an electronics assembly for use in accordance with the present invention; and

FIG. 4 of the drawings is cross-sectional view of a magnetic field generator for use in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

While this invention is susceptible of embodiment in many different forms, there is shown in the drawings and described herein in detail several specific embodiments with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the invention to the embodiments illustrated.

It will be understood that like or analogous elements and/or components, referred to herein, may be identified throughout the drawings by like reference characters. In

addition, it will be understood that the drawings are merely schematic representations of one or more embodiments of the invention, and some of the components may have been distorted from their actual scale for purposes of pictorial clarity.

Referring now to the drawings, and to FIG. 1 in particular, water cooling/recirculating management system 100 is shown which generally comprises controller 10 (See FIG. 3), fluid displacement pump 20, centrifugal separator 30, germicide generator 40, optional continuous, real time corrosion monitoring and reporting system 45, continuous, real time biofilm monitoring and reporting system 47, and primary side stream magnetic field generator 50. Water cooling/recirculating management system 100 eliminates the need for chemicals, bleed water, and labor to maintain recirculating water systems at peak efficiency. It will be understood that, unless otherwise noted, components of the present invention are connected via traditional fluid conduit and/or pipe. As discussed supra, this results in: (1) up to a 40% savings in energy consumption; (2) the elimination of chemical treatment costs; (3) the elimination of blowdown water and sewer costs; (4) the elimination of labor to clean/restore heat exchange equipment; and (5) greatly extended equipment life.

Fluid displacement pump 20 extracts water from main stream basin 60, prefilters the extracted water prior to association with centrifugal separator 30, and displaces the extracted water through a closed-circuit side stream. In accordance with the present invention, fluid displacement pump 20 preferably comprises a one to three horsepower pool pump with a prefilter. Fluid displacement pump 20 is commercially available from, among other pool part vendors, Carvin (Magnum Force Series). Other fluid displacement pumps for use in accordance with the present invention include, for example, U.S. Pat. No. 7,854,597 entitled "Pumping System with Two Way Communication," U.S. Pat. No. 5,730,861 entitled "Swimming Pool Control System," U.S. Pat. No. 5,278,455 entitled "Spa and Pool Pump and Heater Control," and U.S. Pat. No. 4,678,409 entitled "Multiple Magnetic Pump System"—all of which are hereby incorporated herein by reference in their entirety including all references cited therein.

Centrifugal separator 30 includes an inlet, an outlet, and a solids collection chamber. Centrifugal separator 30 separates solids, prevents sludge accumulation, replaces blow-down and/or bleed, removes particles down to 40 microns, contains no filter medium or moving parts, provides for the continuous removal of solids, and/or provides for a purging mechanism that releases solids collected in the collection chamber. Such purging preferably occurs via an actuatable valve in communication with controller 10. Centrifugal separator 30 is commercially available from Lakos Equipment (IL-0150-B), Cole-Parmer, Grainger—among other vendors. Other centrifugal separators suitable for use in accordance with the present invention include, U.S. Pat. No. 8,182,408 entitled "Centrifugal Separator with External Bearings," U.S. Pat. No. 7,927,395 entitled "Centrifugal Separator," U.S. Pat. No. 7,824,459 entitled "Centrifugal Separator," U.S. Pat. No. 7,824,458 entitled "Centrifugal Separator," U.S. Pat. No. 7,811,347 entitled "Centrifugal Separator," U.S. Pat. No. 5,779,619 entitled "Centrifugal Separator," U.S. Pat. No. 5,735,789 entitled "Centrifugal Separator," U.S. Pat. No. 5,591,340 entitled "Centrifugal Separator," United States Patent Application Publication Number 2010/0011723 entitled "Centrifugal Separator," United States Patent Application Publication Number 2009/0013658 entitled "Centrifugal Separator," and Chinese Pat-

ent Number 2355771 entitled "Centrifugal Purifier"—all of which are hereby incorporated herein by reference in their entirety including all references cited therein.

Germicide generator 40 preferably comprises one or more electrodes fabricated from approximately 93 percent copper and approximately 7 percent silver. In another embodiment of the present invention, germicide generator 40 preferably comprises one or more electrodes fabricated from approximately 90 percent copper and approximately 10 percent silver. Germicide generator 40 eliminates the need for chemicals, utilizes a copper/silver ionization system, and destroys algae, bacteria, and viruses. Germicide generator 40 preferably operates one of three modes, namely: an off mode, a continuous on mode, and a non-continuous, pulse mode. Germicide generator 40 is commercially available from Caribbean Clear (ST-1). Other germicide generators suitable for use in accordance with the present invention include, for example, U.S. Pat. No. 9,695,073 entitled "Dual Biocide Generator," U.S. Pat. No. 6,929,740 entitled "Water Purification System," U.S. Pat. No. 5,453,207 entitled "Biocide Treatment System and Method," and International Patent Application Publication Number WO/2007060466 entitled "Ion Generator for Water Treatment"—all of which are hereby incorporated herein by reference in their entirety including all references cited therein.

Continuous, real time corrosion monitoring and reporting system 45 is commercially available from Metal Samples Company (Model MS2100L). Other continuous, real time corrosion monitoring and reporting systems suitable for use in accordance with the present invention include, for example, U.S. Pat. No. 7,239,156 entitled "Configurable Corrosion Measurement Field Device," U.S. Pat. No. 7,245,132 entitled "Intrinsically Safe Corrosion Measurement and History Logging Field Device," U.S. Pat. No. 7,265,559 entitled "Self-Calibrating Corrosion Measurement Field Device with Improved Signal Measurement and Excitation Circuitry," and U.S. Pat. No. 7,282,928 entitled "Corrosion Measurement Field Device with Improved LPE, HDA, and ECN Capability"—all of which are hereby incorporated herein by reference in their entirety including all references cited therein.

In accordance with the present invention optional continuous, real time corrosion monitoring and reporting system 45 preferably comprises a compact corrosion transmitter used in the water treatment industry that measures general corrosion, localized corrosion (pitting), and conductance, and transmits that information to operators via 4 to 20 mA/Highway Addressable Remote Transducer (HART) protocol in real time. The corrosion rate or pitting factor is configured as the primary variable using a standard 2-wire 4 to 20 mA output. The remaining outputs are configured as secondary and tertiary HART variables. Continuous, real time corrosion monitoring and reporting system 45 utilizes state-of-the-art algorithms and data analysis techniques to accurately measure general corrosion rate and pitting. Harmonic distortion analysis (HDA) is applied to improve the performance of the industry-accepted linear polarization resistance (LPR) technique used to measure corrosion rate. To further enhance the performance, an application-specific Stern-Geary variable (B value) is calculated and updated every measuring cycle. There is no need to manually update the B value because of process changes. During the measurement cycle, continuous, real time corrosion monitoring and reporting system 45 also performs an automated electrochemical noise (ECN) measurement, which in combination with the corrosion rate data can provide a measurement of localized corrosion (pitting). Preferably, continuous, real

time corrosion monitoring and reporting system **45** works with three-electrode CorrTran style probes and electrodes, such as, a 3-Electrode Linear Polarization Resistance (LPR) Monitoring. Continuous, real time corrosion monitoring and reporting system **45** can be associated with the mainstream, the side stream or both streams.

Continuous, real time biofilm monitoring and reporting system **47** is commercially available from Metal Samples Company (BG4) and/or Alvim (Italy-BMS). Other continuous, real time biofilm monitoring and reporting systems suitable for use in accordance with the present invention include, for example, U.S. Pat. No. 11,105,730 entitled "System and Method for Detecting Biofilm Growth in Water Systems," and U.S. Pat. No. 9,752,175 entitled "Systems and Methods to Detect Biofilm Streamer Growth and Their Uses"—all of which are hereby incorporated herein by reference in their entirety including all references cited therein.

Continuous, real time biofilm monitoring and reporting system **47** offers a novel approach to monitor the effectiveness of a biocide treatment program to ensure sufficient application of biocide for system cleanliness. It mitigates MIC through real time biofilm monitoring, reduces thermal fouling by controlling biofilms, and optimizes biocide treatment programs. System **47** monitors biofilm activity on the surface of a stainless steel or titanium probe continuously and in real time. In one embodiment, the probe measures cathodic current density in accordance with the teachings of Pavanello et al. in "Exploiting a New Electrochemical Sensor for Biofilm Monitoring and Water Treatment Optimization, Water Research 45 (2011), pp. 1651-1658—all of which is hereby incorporated herein by reference in its entirety including all references cited therein. Continuous, real time biofilm monitoring and reporting system **47** can be associated with the mainstream, the side stream or both streams.

Primary side stream magnetic field generator **50** preferably comprises at least one (e.g. 2, 4, 6, 8, 10, 12, etcetera) permanent ceramic magnet. In accordance with the present invention, the permanent ceramic magnet preferably includes side-to-side polarity that is mounted to a non-ferrous pipe having the south pole facing inward toward the water (See FIG. 4). Primary side stream magnetic field generator **50** prevents scale formation, de-scales existing surfaces and piping, inhibits both general and galvanic corrosion, and, if present, increases heat exchanger efficiency by approximately 5-55 percent. Primary side stream magnetic field generator **50** will be, at the time of publication, commercially available from Prestige Energy Products (Rockford, Michigan).

In accordance with the present invention, side stream magnetic field generator **50** is placed after any turbulent event, such as a pump, filter, phase change, etcetera, and one is also placed just before the cooling tower. The magnetic effect under these circumstances will last roughly 7 minutes. If the pipe length and cycle flow time exceed 7 minutes, then additional magnetic field generators are added.

In further accordance with the present invention, water cooling/recirculating management system **100** preferably includes primary main stream magnetic field generator **70**, secondary main stream magnetic field generator **80**, and/or tertiary main stream magnetic field generator **90**. It will be understood that the main stream magnetic field generators are configured in an analogous manner to the side stream magnetic field generators.

Referring now to FIG. 2, and in an alternative embodiment, water cooling/recirculating management system **100** may also include one or more side stream auxiliary compo-

nents **95**, such as an ultra violet light source, an ozone generator, a pressure switch to indicate the presence of side stream water flow, and/or a cooling fan. Non-limiting examples of auxiliary components suitable for use in accordance with the present invention include, for example, U.S. Pat. No. 9,943,618 entitled "Ultraviolet Light Source and Methods," U.S. Pat. No. 7,683,344 entitled "In-Line Treatment of Liquids and Gases by Light Irradiation," U.S. Pat. No. 6,299,761 entitled "Water Purifier System," U.S. Pat. No. 5,935,431 entitled "Ultraviolet Ozone Water Purifier for Water Disinfection," U.S. Pat. No. 10,724,515 entitled "Proportioning Pump Control Systems and Applicator Apparatus," and U.S. Pat. No. 10,851,638 entitled "Well Fracturing Systems with Electrical Motors and Methods of Use"—all of which are hereby incorporated herein by reference in their entirety including all references cited therein.

As is best shown schematically in FIG. 3, water cooling/recirculating management system **100** includes controller **10**. Controller **10** is in electrical communication with fluid displacement pump **20**, a dump or purge valve associated with centrifugal separator **30**, germicide generator **40**, and continuous, real time corrosion monitoring and reporting system **45**. Controller **10** is optionally in communication with any auxiliary components **95** (e.g., UV light sources, ozone generators, switches, valves, cooling fans, etcetera). While the side stream and primary stream magnetic field generators of the present invention utilize permanent magnets, controller **10** can be in electrical communication if non-permanent and/or electromagnetic components are utilized.

Controller **10** preferably includes a programmable logic controller. The programmable logic controller includes one or more of a battery, a bridge rectifier, a capacitor, a central processing unit, a communications port, a control board, a crystal, a diode, a fuse, a graphics board, an inductor, an input port, an integrated circuit, a microprocessor, a memory module, an oscillator, an output port a potentiometer, a receiver, a relay, a resistor, a semiconductor, a transformer, a transistor, a tuner, a video processing unit, a wired communications hub, and/or a wireless communications hub.

In operation, the present invention is also directed to a method for treating cooling/recirculating water. The method includes extracting water from the main stream basin with the side stream fluid displacement pump. Next, the water is treated (e.g., solids are removed) via the centrifugal separator. The collected solids are periodically discharged via a dump or purge valve. After particle or solid separation, the water is treated with ions (e.g., Cu^{2+} , Ag^{+} , etcetera) via the germicide generator. Next, the water is exposed to at least one side stream magnetic field generator prior to being returned to the main stream basin. Optionally, water from the main stream basin can be treated with one or more main stream magnetic field generators as the water travels through the main stream loop. Notably, the water treatment methods of the present invention utilize physics instead of environmentally problematic chemistry to treat cooling/recirculating water and maintain the system at peak efficiency without the introduction of traditional chemicals.

The foregoing description merely explains and illustrates the invention and the invention is not limited thereto except insofar as the appended claims are so limited, as those skilled in the art who have the disclosure before them will be able to make modifications without departing from the scope of the invention.

While certain embodiments have been illustrated and described, it should be understood that changes and modifications can be made therein in accordance with ordinary

11

skill in the art without departing from the technology in its broader aspects as defined in the following claims.

The embodiments, illustratively described herein may suitably be practiced in the absence of any element or elements, limitation or limitations, not specifically disclosed herein. Thus, for example, the terms “comprising,” “including,” “containing,” etcetera shall be read expansively and without limitation. Additionally, the terms and expressions employed herein have been used as terms of description and not of limitation, and there is no intention in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the claimed technology. Additionally, the phrase “consisting essentially of” will be understood to include those elements specifically recited and those additional elements that do not materially affect the basic and novel characteristics of the claimed technology. The phrase “consisting of” excludes any element not specified.

The present disclosure is not to be limited in terms of the particular embodiments described in this application. Many modifications and variations can be made without departing from its spirit and scope, as will be apparent to those skilled in the art. Functionally equivalent methods and compositions within the scope of the disclosure, in addition to those enumerated herein, will be apparent to those skilled in the art from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims. The present disclosure is to be limited only by the terms of the appended claims, along with the full scope of equivalents to which such claims are entitled. It is to be understood that this disclosure is not limited to particular methods, reagents, compounds compositions or biological systems, which can of course vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

In addition, where features or aspects of the disclosure are described in terms of Markush groups, those skilled in the art will recognize that the disclosure is also thereby described in terms of any individual member or subgroup of members of the Markush group.

As will be understood by one skilled in the art, for any and all purposes, particularly in terms of providing a written description, all ranges disclosed herein also encompass any and all possible subranges and combinations of subranges thereof. Any listed range can be easily recognized as sufficiently describing and enabling the same range being broken down into at least equal halves, thirds, quarters, fifths, tenths, etcetera. As a non-limiting example, each range discussed herein can be readily broken down into a lower third, middle third and upper third, etcetera. As will also be understood by one skilled in the art all language such as “up to,” “at least,” “greater than,” “less than,” and the like, include the number recited and refer to ranges which can be subsequently broken down into subranges as discussed above. Finally, as will be understood by one skilled in the art, a range includes each individual member.

All publications, patent applications, issued patents, and other documents referred to in this specification are herein incorporated by reference as if each individual publication, patent application, issued patent, or other document was specifically and individually indicated to be incorporated by reference in its entirety. Definitions that are contained in text incorporated by reference are excluded to the extent that they contradict definitions in this disclosure.

Other embodiments are set forth in the following claims.

12

What is claimed and desired to be secured by Letters Patent of the United States is:

1. A system, comprising:

- a controller, wherein the controller is in communication with a fluid displacement pump, a centrifugal separator purge valve, and a germicide generator;
- a fluid displacement pump, wherein the fluid displacement pump extracts water from a main stream basin, prefilters the extracted water, and displaces the extracted water through a closed-circuit side stream;
- a centrifugal separator, wherein the centrifugal separator includes an inlet, an outlet, a solids collection chamber, and a purge valve that releases solids collected in the collection chamber;
- a germicide generator, wherein the germicide generator comprises at least one electrode fabricated from approximately 93 percent copper and approximately 7 percent silver;
- a primary side stream magnetic field generator, wherein the primary side stream magnetic field generator comprises at least one permanent ceramic magnet having side-to-side polarity mounted to a non-ferrous pipe with a south pole facing inward toward water;
- a continuous, real time corrosion monitoring and reporting system; and
- a continuous, real time biofilm monitoring and reporting system, wherein the biofilm monitoring system measures cathodic current density on a surface of a probe continuously and in real time.

2. A system, comprising:

- a controller in electrical communication with a fluid displacement pump, a centrifugal separator purge valve, and a germicide generator;
- the fluid displacement pump that extracts water from a main stream basin through a closed-circuit side stream;
- the centrifugal separator having a solids collection chamber with the purge valve;
- the germicide generator comprising at least one electrode fabricated from approximately 93 percent copper and approximately 7 percent silver;
- a primary side stream magnetic field generator comprising at least one permanent ceramic magnet;
- a continuous, real time corrosion monitoring and reporting system that uses linear polarization resistance technique with harmonic distortion analysis; and
- a continuous, real time biofilm monitoring and reporting system that measures cathodic current density on a stainless steel or titanium probe surface, wherein the controller operates the purge valve, controls the germicide generator in at least one of an off mode, continuous on mode, and non-continuous pulse mode, and receives real-time data from both monitoring systems.

3. A system, comprising:

- a controller with programmable logic controller functionality;
- a fluid displacement pump comprising a one to three horsepower pool pump with prefilter;
- a centrifugal separator that removes particles down to 40 microns and contains no filter medium or moving parts;
- a germicide generator comprising electrodes fabricated from approximately 93 percent copper and approximately 7 percent silver operating in a copper/silver ionization system;
- a primary side stream magnetic field generator comprising permanent ceramic magnets with side-to-side polarity

13

mounted to a non-ferrous pipe having a south pole facing inward toward water;

a continuous, real time corrosion monitoring and reporting system utilizing linear polarization resistance with harmonic distortion analysis and electrochemical noise measurement; and

a continuous, real time biofilm monitoring and reporting system measuring cathodic current density on a probe surface continuously, wherein the system eliminates chemical treatment costs, blowdown water, and provides up to 40% energy savings through physics-based water treatment.

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14